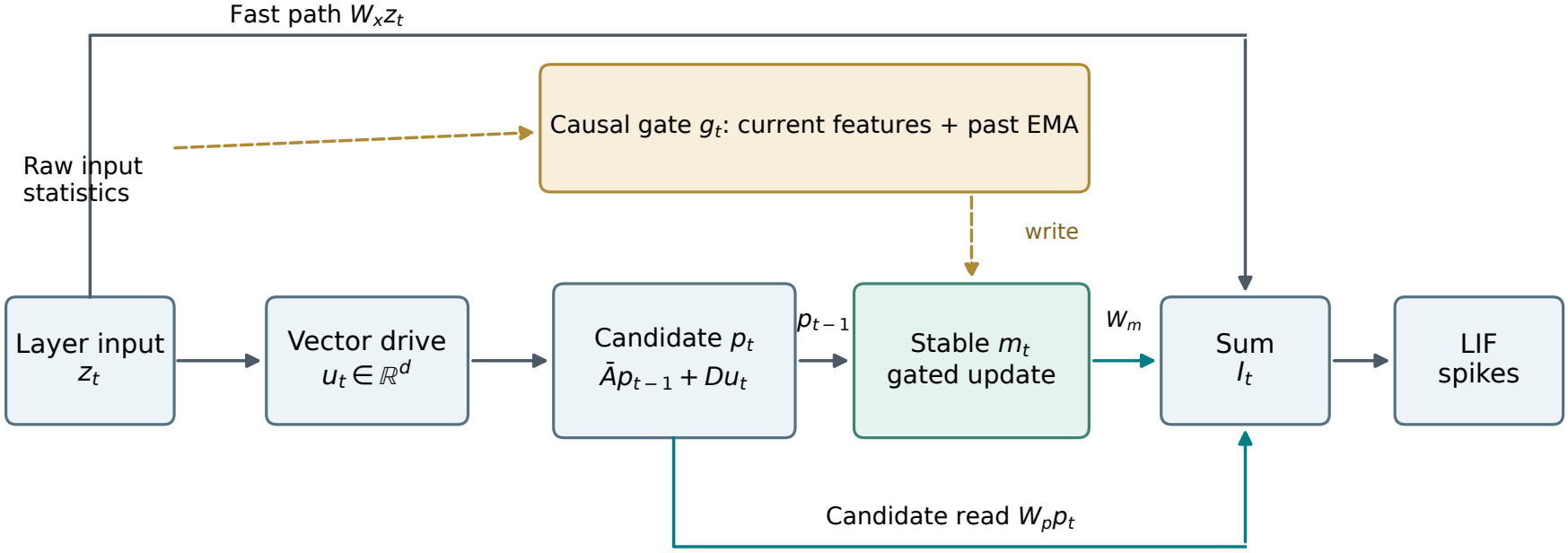

$$m_t = (1 - g_t) \odot m_{t-1} + g_t \odot p_{t-1}$$

When the write gate is closed, stable memory is held; candidate and fast paths keep running.